



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to CHENNAMMAL AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

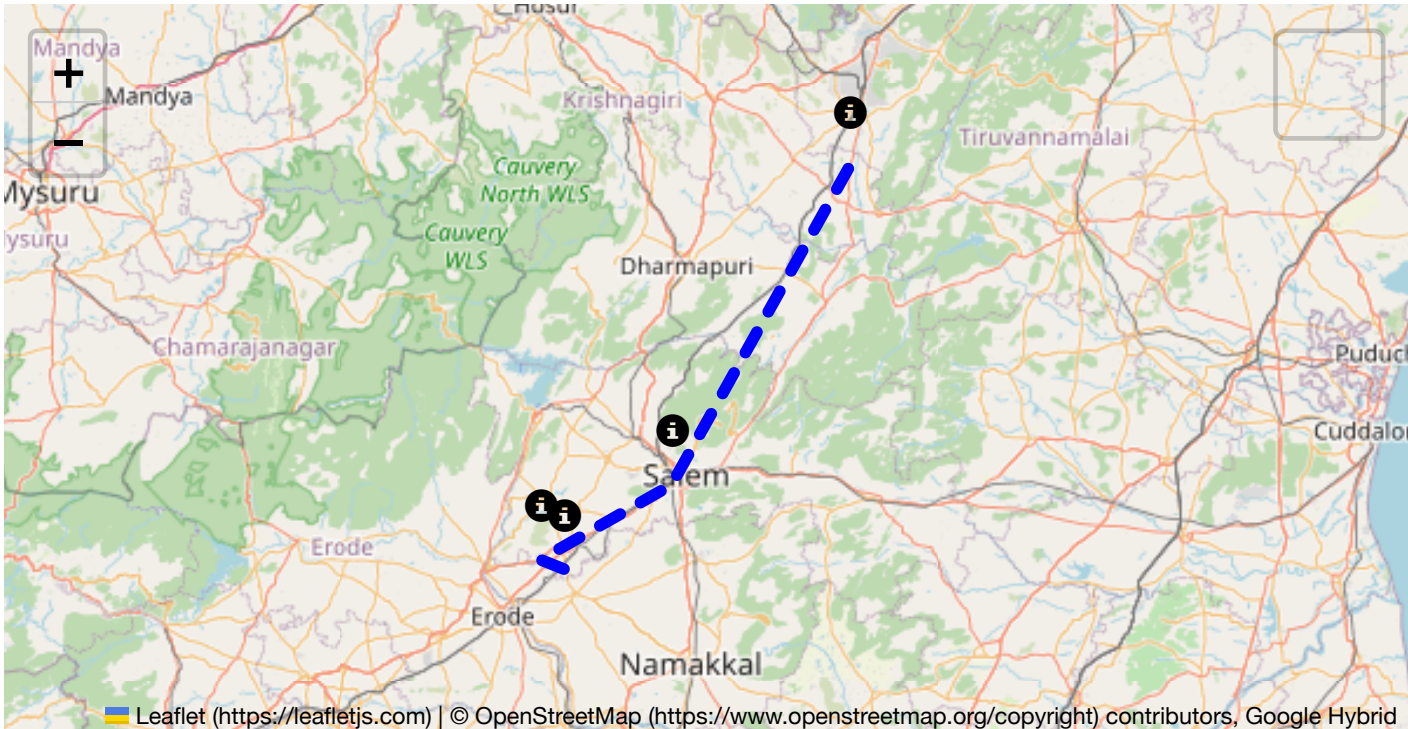
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 156.18 km
Estimated Duration: 3.5 hours
Adjusted Duration (Heavy Vehicle): 4.4 hours

Start: (11.4381, 77.8734)
End: (12.362, 78.541)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

This route begins at CVQF+23W, Sangagiri, Tamil Nadu, progresses through Bhavani Main Road in Padaiveedu, and then passes through Coimbatore Rd in Kondalampatti, Salem, finally reaching 9G6R+Q95 in Kallavi, Tamil Nadu. The route stretches approximately 156.18 kilometers, characterized by a mix of urban and rural landscapes. The journey typically takes about 3.51 hours when accounting for the transport of heavy vehicles carrying hazardous materials.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate with distinct wet and dry seasons. The monsoon season, lasting from June to September, can bring heavy rainfall, potentially causing flooding and slippery road surfaces. During summer months (March to May), high temperatures may lead to road surface issues such as softening asphalt, which can affect vehicle handling. Additionally, occasional dense fog can occur in the cooler months of December and January, reducing visibility significantly.

3. Analysis of Traffic Patterns

- Sangagiri and Salem:** Moderate to heavy traffic, particularly during rush hours (8-10 AM and 5-8 PM). Salem, in particular, may have congestion near market areas and industrial zones.
- Rural Segments:** Lower traffic density, but agricultural vehicles and local commuters can cause slowdowns.
- Congestion-Prone Areas:** Key congestion points include the towns of Sangagiri and Kondalampatti, especially where roads intersect with local businesses and market areas.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** The route encompasses both well-maintained highways and smaller rural roads, which may have potholes or uneven surfaces.
- **Infrastructure:** Expect standard two-lane highways in rural sections and broader roads in urban areas, but watch for narrow sections near villages and potential obstructions from market activity.

5. Suggestions for Alternative Routes for Emergencies

- **Alternate Route A:** A potential diversion could bypass Salem via State Highway 188, though this may add both time and distance.
- **Alternate Route B:** Consider FR48+7XQ to FR54, leading to Kallavi by avoiding some congested urban centers.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials in Tamil Nadu requires adherence to strict regulations, including obtaining necessary permits, using appropriate safety markings, and operating during designated times (usually avoiding nighttime transport unless specifically permitted).

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While data on specific incidents is limited, the region has witnessed some accidents involving heavy vehicles, primarily due to driver fatigue and adverse weather conditions. Notably, intersections and sharp bends in rural areas have been identified as potential risk points.

8. Environmental Considerations and Sensitive Areas

The route crosses near agricultural regions and small villages, necessitating caution to avoid spillages that could affect local water supplies or agricultural fields. Proximity to wildlife reserves should also prompt adherence to speed limits to prevent disturbances.

9. Analysis of Communication Coverage

Mobile networks generally provide good coverage along the route, especially near urban areas. However, certain rural sections may experience intermittent signal loss, particularly around hilly or forested terrain.

10. Estimated Emergency Response Times for Different Route Segments

- **Urban Areas (Salem, Sangagiri):** Emergency services response times can be estimated at approximately 15-30 minutes due to better infrastructure and proximity to service facilities.
- **Rural Sections:** Response times may extend to 45-60 minutes due to access issues and the distance from emergency service hubs.

12. Overall Summary of Risk Assessment

The route poses moderate risk levels chiefly due to variable road conditions, potential for traffic congestion, and environmental factors such as weather. Key risk mitigation steps include thorough route planning to avoid peak hours and preparing for adverse weather; ensuring vehicle maintenance to handle variable road surfaces; and maintaining communication with local authorities and emergency services. Regular checks and adherence to regulations regarding hazardous materials will further ensure a safer transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45353, 77.85697	15 KM/Hr
5	Turn	Medium	11.45781, 77.85203	30 KM/Hr
6	Turn	High	11.46307, 77.84941	15 KM/Hr
7	Turn	Medium	11.45546, 77.81800	30 KM/Hr
8	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
9	Turn	High	11.53573, 77.95794	15 KM/Hr
10	Turn	Medium	11.62100, 78.10723	30 KM/Hr
11	Blind Spot	Blind Spot	11.62130, 78.10776	10 KM/Hr
12	Blind Spot	Blind Spot	11.61235, 78.10438	10 KM/Hr
13	Turn	Medium	11.61235, 78.10495	30 KM/Hr
14	Blind Spot	Blind Spot	11.62871, 78.12299	10 KM/Hr
15	Blind Spot	Blind Spot	11.62788, 78.12219	10 KM/Hr
16	Turn	High	11.63302, 78.12742	15 KM/Hr
17	Turn	Medium	11.62938, 78.16601	30 KM/Hr
18	Turn	Medium	11.65825, 78.20469	30 KM/Hr
19	Turn	Medium	11.66366, 78.20649	30 KM/Hr
20	Turn	High	11.67116, 78.23102	15 KM/Hr
21	Turn	Medium	11.67182, 78.23141	30 KM/Hr
22	Turn	Medium	11.67500, 78.23962	30 KM/Hr
23	Turn	Medium	11.80474, 78.33353	30 KM/Hr
24	Turn	Medium	11.81272, 78.34092	30 KM/Hr
25	Turn	Medium	11.83514, 78.35305	30 KM/Hr
26	Turn	Medium	11.99608, 78.44589	30 KM/Hr
27	Turn	Medium	12.04286, 78.48352	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
28	Turn	Medium	12.04845, 78.48385	30 KM/Hr
29	Turn	Medium	12.04958, 78.48314	30 KM/Hr
30	Turn	Medium	12.06149, 78.47722	30 KM/Hr
31	Turn	Medium	12.06932, 78.47655	30 KM/Hr
32	Turn	Medium	12.07112, 78.47724	30 KM/Hr
33	Turn	Medium	12.11669, 78.49766	30 KM/Hr
34	Turn	Medium	12.17764, 78.54800	30 KM/Hr
35	Turn	Medium	12.17847, 78.54856	30 KM/Hr
36	Turn	Medium	12.18223, 78.54843	30 KM/Hr
37	Turn	High	12.18614, 78.54962	15 KM/Hr
38	Turn	Medium	12.20953, 78.54358	30 KM/Hr
39	Turn	High	12.21030, 78.54402	15 KM/Hr
40	Turn	Medium	12.25956, 78.53301	30 KM/Hr
41	Turn	Medium	12.26092, 78.53324	30 KM/Hr
42	Turn	Medium	12.26207, 78.53436	30 KM/Hr
43	Turn	Medium	12.31095, 78.53818	30 KM/Hr
44	Turn	High	12.36356, 78.54371	15 KM/Hr
45	Turn	High	12.36375, 78.54324	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
4	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
7	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
11	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
12	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
14	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
15	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
16	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
17	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
18	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
19	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
20	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
21	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
22	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
23	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
24	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
26	hospital	Dr.Thiru Neuro Multi Speciality	11.6623376, 78.203353	30 km/h	Medium
28	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
29	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
32	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium
37	hospital	Nanjaya Hospital	12.0550888, 78.4821929	30 km/h	Medium
38	hospital	Subam Hospital	12.0595128, 78.4766713	30 km/h	Medium
39	hospital	Government Hospital, Harur	12.061178, 78.4756913	30 km/h	Medium
40	hospital	Government Hospital, Uthangarai	12.26612, 78.535949	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
6	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
9	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
10	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
13	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
25	school	Holy Cross Matriculation Higher Secondary School	11.6635657, 78.203532	30 km/h	Medium
27	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
30	school	Indian Matriculation Higher secondary school	12.006299, 78.4531204	30 km/h	Medium
31	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
33	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
34	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
35	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
36	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.63266, 78.12699



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.53573, 77.95794



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62100, 78.10723



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61235, 78.10438



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61235, 78.10495



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62871, 78.12299



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62788, 78.12219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63302, 78.12742



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62938, 78.16601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65825, 78.20469



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66366, 78.20649



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67116, 78.23102



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67182, 78.23141



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67500, 78.23962



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80474, 78.33353



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81272, 78.34092



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83514, 78.35305



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99608, 78.44589



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04286, 78.48352



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04845, 78.48385



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04958, 78.48314



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.06149, 78.47722



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.06932, 78.47655



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.07112, 78.47724



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11669, 78.49766



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17764, 78.54800



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17847, 78.54856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.18223, 78.54843



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.18614, 78.54962



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20953, 78.54358



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21030, 78.54402



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25956, 78.53301



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26092, 78.53324



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26207, 78.53436



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31095, 78.53818



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.36356, 78.54371



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.36375, 78.54324

